AQA

Please write clearly in block capitals.

Centre number | | | | | | Candidate number | | | | |

Surname _____

Forename(s) _____

Candidate signature _____

I declare this is my own work.

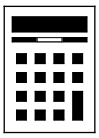
Functional Skills Level 1
MATHEMATICS

Paper 2 Calculator

Thursday 3 November 2022 Afternoon Time allowed: 1 hour 30 minutes

Materials

For this paper you must have:
- a calculator
- mathematical instruments.



For Examiner's Use	
Question	Mark
1–8	
9	
10	
11	
12	
TOTAL	

Instructions
- Use black ink or black ball-point pen. Draw diagrams in pencil.
- Answer **all** questions.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- If you need extra space for your answer(s), use the lined pages at the end of this book. Write the question number against your answer(s).
- Do all rough work in this book. Cross through any work you do not want to be marked.
- State the units of your answer where appropriate.

Information
- The marks for questions are shown in brackets.
- The maximum mark for this paper is 60.
- You may ask for more answer paper, graph paper and tracing paper. These must be tagged securely to this answer book.

Advice

In all calculations, show clearly how you work out your answer.

Do not write outside the box

Section A

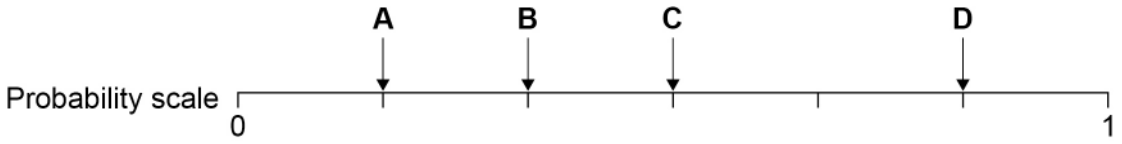
Answer **all** questions in the spaces provided.

1 A fair six-sided dice is rolled.

Which letter shows the probability of landing on an even number?

Circle your answer.

[1 mark]



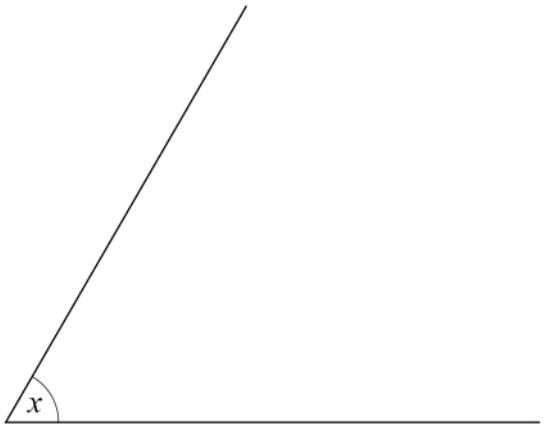
2 Circle the fraction with the same value as $3\frac{3}{4}$

[1 mark]

$\frac{9}{4}$ $\frac{10}{4}$ $\frac{13}{4}$ $\frac{15}{4}$

3 Measure the size of angle x

[1 mark]



Answer _____ °

Do not write outside the box

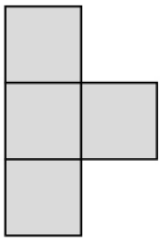
4 Calculate -5×-6

[1 mark]

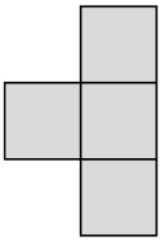
Answer _____

5 The front elevation and side elevation of a shape are shown.

FRONT ELEVATION



SIDE ELEVATION

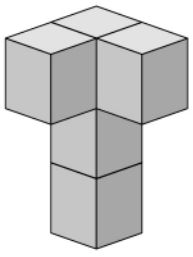


Which shape has these elevations?

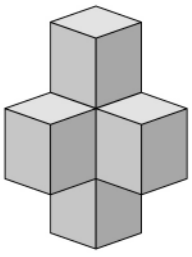
Circle the correct letter.

[1 mark]

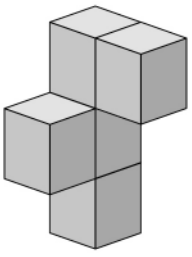
A



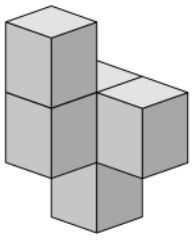
B



C



D



Do not write outside the box

6 Round 14.76 to one decimal place.

[1 mark]

Answer _____

7 Here are six numbers.

16 19 16 10 17 12

Work out the mean.

[3 marks]

_____

_____

_____

_____

Answer _____

8 Increase 630 by 40%

[3 marks]

_____

_____

_____

_____

Answer _____

12

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Section B

Answer **all** questions in the spaces provided.

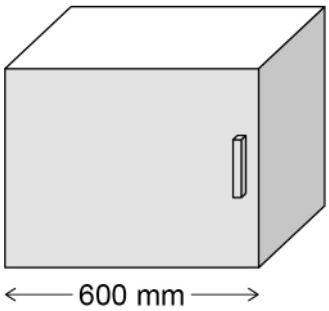
9 **The Kitchen Fitter**

Ryan is a kitchen fitter.

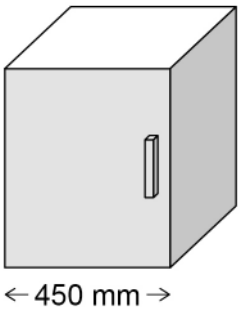
9 (a) A customer wants cupboards along one wall of her kitchen.

Ryan can choose from these cupboards.

Large



Small



He can choose some of each size.

The length of the wall is 3.45 **metres**.

How many of each cupboard could Ryan choose so that they fit the length of the wall with no gaps?

[4 marks]

Number of large cupboards _____

Number of small cupboards _____

Question 9 continues on the next page

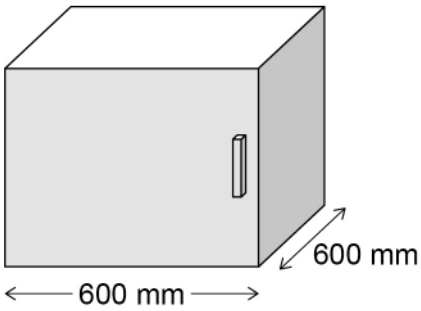
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9 (b) A different customer wants

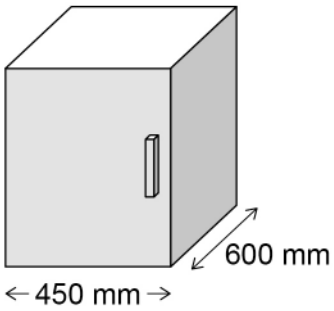
3 large cupboards with width 600 mm and depth 600 mm

2 small cupboards with width 450 mm and depth 600 mm

Large



Small



The cupboards will fit along **one** wall of this customer's kitchen.

The grid on the page opposite shows a scale drawing of the kitchen.

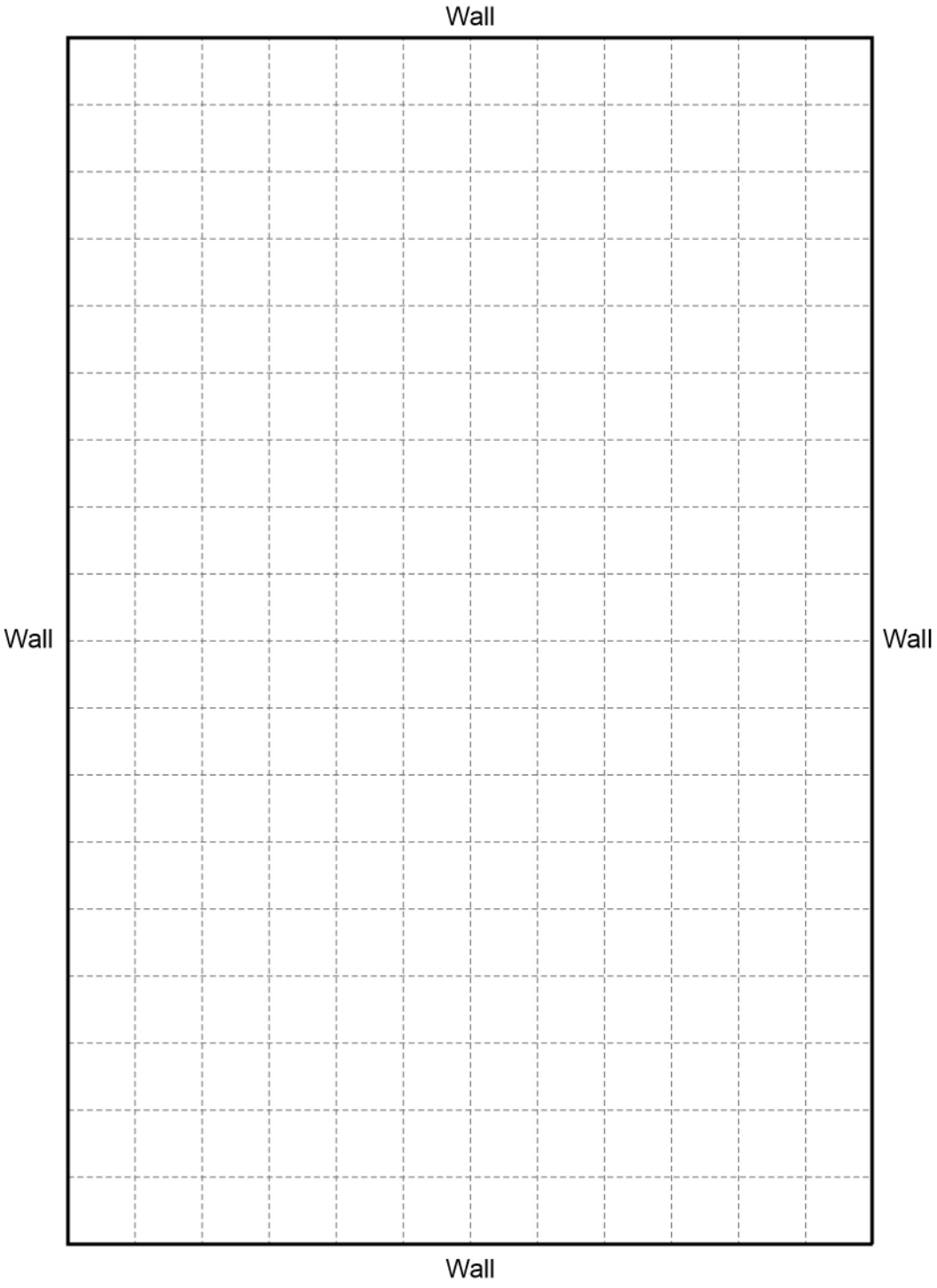
The scale is 1 cm represents 150 mm

Use the grid to draw a plan of where Ryan could fit the cupboards.

[4 marks]

Do not write outside the box

Scale: 1 cm represents 150 mm



Question 9 continues on the next page

Do not write outside the box

9 (c) This is the price list for Ryan's kitchen cupboards.

	Price per cupboard
Large cupboard	£85.50
Small cupboard	£75.00

There is a special offer.

Large cupboards
Buy one, get one half price

A customer wants to order

6 large cupboards

and

2 small cupboards.

The customer has a budget of £550

Does the customer have enough money to buy these cupboards?

You **must** show your working.

[5 marks]

13

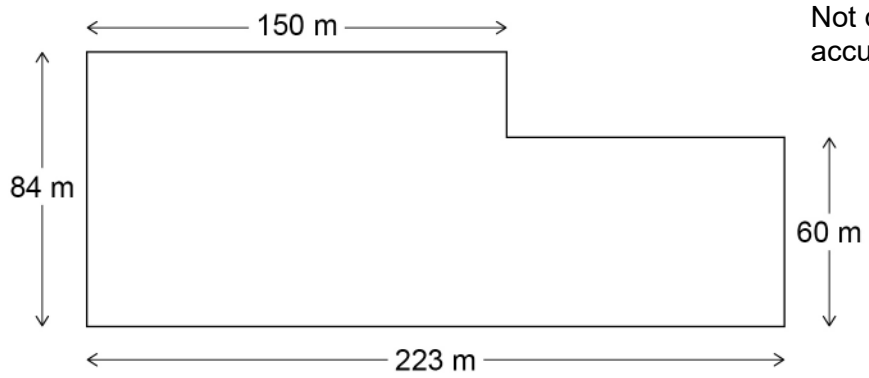
Do not write outside the box

10 Flower Farm

Erik grows daffodils on his farm.



10 (a) This is a sketch of a field where Erik will grow daffodils.



Not drawn accurately

Erik plants 1.25 kg of daffodil bulbs per square metre of land.

He has 20 tonnes of bulbs.

1 tonne = 1000 kg

Does Erik have enough bulbs to plant all of this field?

You **must** show your working.

[5 marks]

Question 10 continues on the next page

Do not write outside the box

10 (b) Joe picks flowers on Erik's farm.

Joe is paid £8.29 for each box he fills with flowers.

The flower-picking season lasts for 10 weeks.

The list shows the number of boxes Joe filled each week for the first 5 weeks.

55 47 45 44 59

Estimate how much Joe can expect to earn for the full 10-week season.

You **must** show your working.

[4 marks]

Answer £ _____

Do not write outside the box

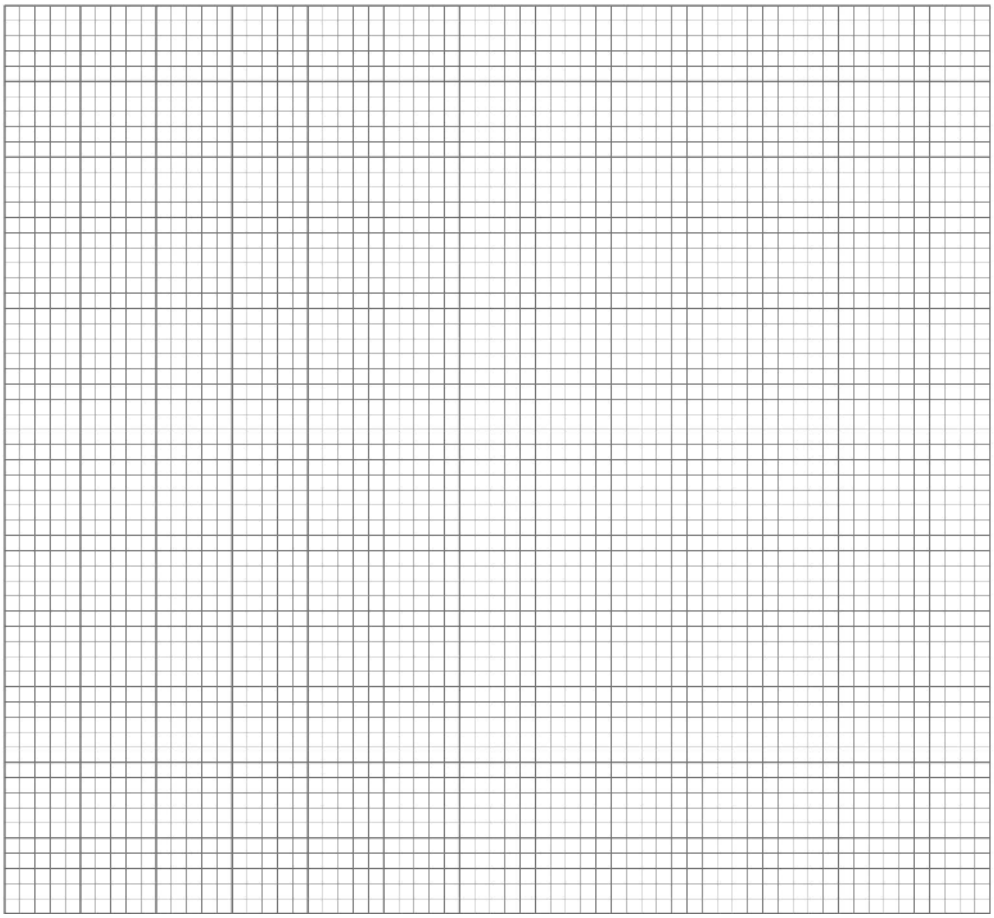
10 (c) Erik plants different types of daffodils in his fields.

Here is some data about the area he planted with each type of daffodil.

Type of daffodil	Area planted (m^2)
Golden Ducat	88 000
White Lion	72 000
Ice Follies	46 000
Pink Pride	34 000

Draw a suitable chart to show this data.

[4 marks]



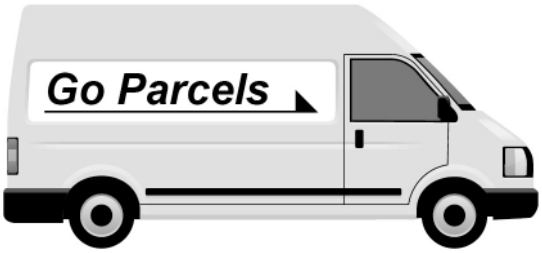
13

Turn over for the next question

Do not write outside the box

11 **Parcels**

Sam sends parcels using couriers.



11 (a) The table shows information about the cost of sending parcels with Go Parcels.

Weight range	Cost
Up to 1 kg	£5.99
More than 1 kg up to 10 kg	£12.99
More than 10 kg up to 15 kg	£14.24
More than 15 kg up to 20 kg	£25.49

Sam needs to send items weighing 2 kg, 4 kg and 11 kg

Sam can combine the three items into one parcel before sending.

How much cheaper is it to send one combined parcel rather than three separate parcels?

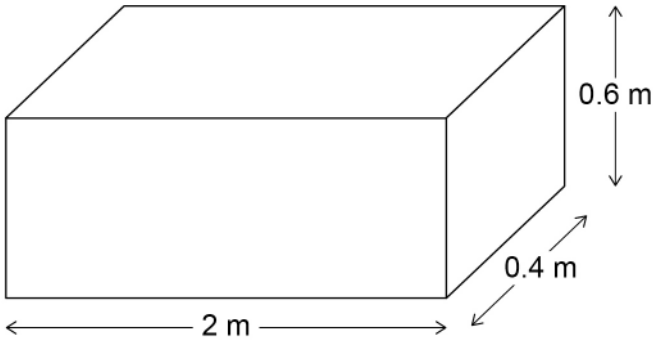
[4 marks]

Answer £ _____

Do not write outside the box

11 (b) Sam sends a parcel with a different courier, Package Plus.

The parcel is a cuboid box measuring 2 m by 0.4 m by 0.6 m



Package Plus charges using the formula

$$\text{Cost (£)} = £15.25 + (\text{volume of parcel in m}^3 \times £4.25)$$

Sam pays with a £20 note.

How much change should Sam receive?

[5 marks]

Answer £ _____

9

Turn over ►

Do not write outside the box

12 **The Theatre**

Amy is going to a theatre show in Chichester.

12 (a) Amy will travel by train to Chichester.

Amy has a railcard that gives her $\frac{1}{3}$ off the normal price of a train ticket.

The normal price of a return ticket to Chichester is £16.83

Amy says,

"My ticket will cost **less than** £11 if I use my railcard."

Is Amy correct?

You **must** show your working.

[3 marks]

Do not write outside the box

12 (b) The theatre sells ice creams at the interval.

All the ice creams cost the same amount.

Amy buys herself and her two friends an ice cream each.

The total cost is £8.40

How much does it cost to buy 7 ice creams?

[3 marks]

Answer £ _____

Question 12 continues on the next page

Do not write outside the box

12 (c) At the theatre, the show starts at 7.30 pm

The show lasts for 105 minutes plus a 20-minute interval.

At the end of the show, it will take Amy 35 minutes to get to the station.

A train leaves the station at 22.17

Can Amy catch this train?

You **must** show your working.

[4 marks]

Do not write outside the box

12 (d) Jai works in the box office at the theatre.

The theatre has 1206 seats.

For this performance, the ratio of tickets sold to tickets not sold is 5 : 1

Jai says,

"We have sold more than 1000 tickets for this performance."

Show that he is correct.

[3 marks]

13

END OF QUESTIONS

Do not write outside the box

There are no questions printed on this page

DO NOT WRITE ON THIS PAGE
ANSWER IN THE SPACES PROVIDED

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Question number	**Additional page, if required.** **Write the question numbers in the left-hand margin.**

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Question number	**Additional page, if required.** **Write the question numbers in the left-hand margin.**

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